



ATMOS 2026 notification for paper 7

À partir de ATMOS 2026 <atmos2026@easychair.org>

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À BAUDRU Julien <Julien.Baudru@ulb.be>

We are sorry to inform you that your paper

Exploring Multi-Order Markov Chains for Road Network Mobility Prediction

has not been accepted for presentation at ATMOS 2026.

We received a record high number of submissions this year, which made competition for the presentation slots fierce, and some strong submissions could not be accommodated in the final program.

Thank you for submitting your work to ATMOS 2026.

ATMOS 2026 PC chairs

Valentina Cacchiani and Stefan Funke

SUBMISSION: 7

TITLE: Exploring Multi-Order Markov Chains for Road Network Mobility Prediction

----- REVIEW 1 -----

SUBMISSION: 7

TITLE: Exploring Multi-Order Markov Chains for Road Network Mobility Prediction

----- Overall evaluation -----

SCORE: 0 (borderline paper)

----- TEXT:

SUMMARY

The paper addresses the problem of predicting the next node of a trajectory on a road network. The motivation is a gap between deep learning models, which are accurate but expensive to train and unable to adapt while a trip is in progress, and simpler statistical models, which are cheap but require delicate design choices. The first contribution is a family of three predictors of increasing complexity, each adding one ingredient: the first uses only how far back the past trajectory is taken into account, the second combines several such choices instead of picking one, and the third additionally keeps a record of the individual user's recent movements alongside the patterns learned from the whole population.

The second contribution is evaluative. Besides the standard task of predicting one step ahead with the true past always available, the authors propose a harder task in which the model is given only a short initial fragment and must then continue on its own, feeding its own predictions back as input; performance is measured by how far it can go before accuracy degrades and by how close it ends up to the true destination. Experiments are run on four road networks taken from OpenStreetMap,

using synthetic trajectories mapped onto them and released as a benchmark, and compared against established methods.

The main finding is that the three variants are almost indistinguishable in accuracy and match or beat the existing methods, but differ substantially in how confident they are: the variant with individual memory places much more probability on the node that actually occurs. The authors also acknowledge that road networks offer few choices at each junction, which limits how much the models can differ from one another.

EVALUATION

- Novelty/originality. The individual ingredients are all known in the sequence-prediction literature; the originality lies in the systematic decomposition and in the evaluation protocol rather than in the models themselves, so the novelty is modest, though consistent with the short-paper format.
- Impact on applications of interest to ATMOS. The work relates pertinently to Intelligent Transportation Systems, Models for User Behaviour and Demand Forecasting, and the second evaluation task has a concrete operational reading (how long a system can be left to forecast on its own between sensor or tracking updates) but this applicative connection remains a passing remark rather than a developed use case.
- Theoretical/algorithmic impact with respect to ATMOS topics. Here the contribution is essentially absent: no computational problem is studied, there is no pseudocode, no complexity analysis, no guarantees on the behaviour of the estimator, and most critically the central claim of "lightweightness" is never verified, since no running times, memory footprints or construction costs are reported. The parameter choices also lack any criterion or sensitivity analysis, with automatic tuning deferred to future work.
- Presentation. The text is clear, well organised and readable, though I found the formalism of equations (2) and (4) dense and presented without accompanying intuition. The results table would have benefited from highlighting the best values.
- Positioning w.r.t the state of the art. The comparison is properly conducted against relevant baselines, with reimplementations for fairness, but it is weak since there is no comparison against the deep learning models cited in the introduction as the motivation for the entire work. Moreover, the evaluation is carried out entirely on synthetic trajectories, so it remains unknown how the reported accuracy would hold up on real mobility data.

----- REVIEW 2 -----

SUBMISSION: 7

TITLE: Exploring Multi-Order Markov Chains for Road Network Mobility Prediction

----- Overall evaluation -----

SCORE: -1 (weak reject)

----- TEXT:

The aim of the paper is to predict the node a traveler in a road network will visit next. The authors propose to use a Markov chain for this, where the prediction for the next node depends on the 10 previously visited nodes. The authors provide three variants for the Markov Chain model and evaluate them in two settings: In the first setting, the ground-truth path is revealed one node at a time and the model has to predict the next node at each time step. In the second variant, the model only gets the first 10 nodes of the trajectory and then predicts the full trajectory step by step without getting additional information (i.e., later predictions do not only depend on the ground truth but also on earlier predictions). The experiments indicate that the prediction quality is competitive with more elaborate (and computationally expensive) models.

The paper seems generally interesting, but I have some reservations:

- * The mathematical definitions are sloppy. Parsing them requires guesswork and takes much more time than necessary. This makes it hard to interpret and judge the results. See [1] for an example.

- * The writing seems quite convoluted. There are sentences that sound very impressive at first glance but either have a very simple meaning or are not parsable (at least not by me). Besides being difficult to read, it also again makes it difficult to interpret the results. See [2] for examples.

- * The paper is centered around a question of the form "How can we learn from historical data to predict the future?" rather than "How can we efficiently solve an algorithmic problem?". Thus, while the topic is fitting for ATMOS, the methodological connection is not the best fit to the call for papers [3]. I would not put too much weight on this, as I can see that the insights might be interesting for the algorithms community. However, I would have liked to see a stronger link made in the paper.

With these weaknesses, I lean towards rejection.

[1] In the definitions, many symbols are used without specifying their meaning or even their type. One can guess this from the context, but a proper definition would be appropriate here. Examples from just the first definition (Max-Order Markov Chain):

- * "Given a partial trajectory $(v_1, \dots, v_t) \in V^t$: Is there any restriction on such a tuple to be a trajectory? I assume that each pair of consecutive vertices needs to be connected by an edge? Also the next predicted node needs to be connected by an edge, I assume?

- * "k-gram counts $C_{\{c, v\}}^k$ accumulated over training trajectories": What is this? What is c and v ? I assume c is a given context c^k ? And I guess v is a candidate vertex to come next? If so, I assume that $C_{\{c, v\}}^k$ is the number of times that the vertex v comes after the k -path c in the training data?

- * "with support $N_c^k = \dots$ ": So I guess this is just the number of times the path c of length k appears as a subpath in the training data? Why not say this? Why do we sum over " v " here? Why not just use " v "? Also, why is this called the "support"? Maybe I am missing something, but this seems quite different from what "support" usually means in mathematics.

- * "the model selects the highest order whose context was observed and returns its empirical distribution $P_{MC}(v) = \dots$ ": What? I cannot parse this sentence. The defined $P_{MC}(v)$ seems to be simply the probability that the next vertex of a path is v conditioned on the previous k vertices being c . It is unclear to me, how this probability is now used to make the prediction. Also, the notation seems inconsistent. Why does $P_{MC}(v)$ not depend on c ? Why does it not have a (k) in the exponent as all notions before? What is " k^* "? I assume it is just " k ". A few lines below, $P_{MC}(v)$ seems to be used but with the name $P^k(v)$. What is going on here? Why is it now called differently with an explicit dependence on k ?

[2] Here are just some examples of overly convoluted formulations:

- * In multiple places, a "dual experimental setup" or "dual evaluation protocol" is mentioned, as if this was a technical term referring to something very specific. If I understand it correctly, this simply means that two experiments are done.

* "The dual evaluation protocol comprises two complementary tasks, each reflecting a distinct operational context.": How are the tasks complementary? What does "reflecting a distinct operational context" mean beyond "they are different experiments"?

* "progressively combining adaptive order selection with dual population-individual memory": I do not understand what is being said here.

[3] Methods mentioned in the call for paper: Algorithmic Game Theory, Approximation Algorithms, Combinatorial Optimization, Graph and Network Algorithms, Heuristics and Meta-heuristics, Mathematical Programming, Methods for the Integration of Planning Stages, Online and Real-time Algorithms, Simulation Tools, Stochastic and Robust Optimization

----- REVIEW 3 -----

SUBMISSION: 7

TITLE: Exploring Multi-Order Markov Chains for Road Network Mobility Prediction

----- Overall evaluation -----

SCORE: 0 (borderline paper)

----- TEXT:

The submission studies next-node trajectory prediction on road networks using a family of Markov chain models of increasing complexity: A maximum-context variant, an entropy-weighted variant aggregating over all observed contexts, and a variant that combines population-level patterns with individual user history.

They are evaluated for next-node predictions and for auto-regressive forecasting over longer horizons with comparisons against four existing approaches.

The topic is relevant to the scope of ATMOS.

The main contributions are the family of Markov chain models along with their evaluation and an adaptation of WorldMove mapped onto OpenStreetMap road networks, released as an open benchmark.

My recommendation is to consider the submission as borderline:

It is clearly written, and the WorldMove-to-OSM benchmark and the accompanying code are actual contributions.

On the other hand, the modelling novelty is thin and several empirical claims are not convincing.

The core ideas shown in the submission are well established.

Higher- and variable-order Markov models, entropy-based order weighting, Dirichlet smoothing, and population/individual memory blending are all standard constructions that the authors combine rather than advance.

The weighting of population-level patterns with user history is the most interesting element, but as a heuristic with several parameters, it could profit from more analysis in the future.

The empirical claims overstate the numbers.

On next-node accuracy, the stronger baselines and the top variants from the submission all perform similarly, well within one reported standard deviation, so claiming "outperform" is a stretch.

The headline result — that weighting population-level data with individual user data produces an order-of-magnitude gain in certainty — is surprising, but also concerning.

The authors should validate that these are true findings and not an artifact of the methodology or a bug.

Is the maximum-context variant unsmoothed (assigning near-zero probability to unseen successors)? This could explain much the order-of-magnitude gap.

The evaluation also rests on a single synthetic source, which could impact results since the algorithm leverages recurrent individual-level regularities that synthetic users are less likely to reproduce faithfully.

Looking at the auto-regressive predictions, one could have also assumed that a higher level of certainty would lead to better results, but the most straightforward approach was able to predict the most nodes.

The framing of the submission also calls out a focus on lightweight/real-time deployments.

Evidence to that end should be given more specifically in the experimental section including a comparison to the baselines to underpin the stated motivation.

Minor Notes:

- p2: Parameter k -star is not introduced. The preceding sentence should probably read "...selects the highest order k -star...".
- p2: Notation-wise, keeping the (k) , (k^*) superscripts with context c reads clearer than moving it to C and N .
- p3: Parameters β and V are not introduced. In the preceding sentence, just state that they are for Dirichlet smoothing.
- p3: Many hyperparameters (τ , λ , ϵ , β , η , p , κ , N) are introduced without values or a selection procedure. While a full sensitivity analysis is not needed for a short paper, a short note on how they were set would help reproducibility and boost reader confidence.
- p5: In Table 2 the MC2 and HW-MC2 rows are identical across every metric. Please verify if accurate and if yes, note whether this is an effect of the short context.
- p5: The readability of Table 2 would benefit from showing fewer digits. Also, stating 1 - Accuracy values would highlight differences better.